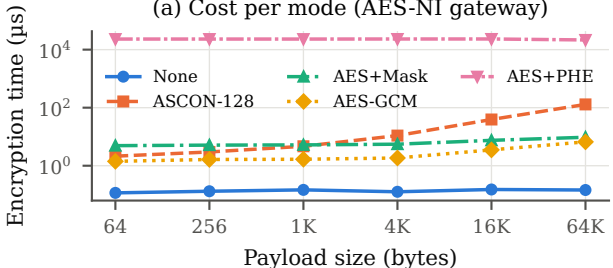
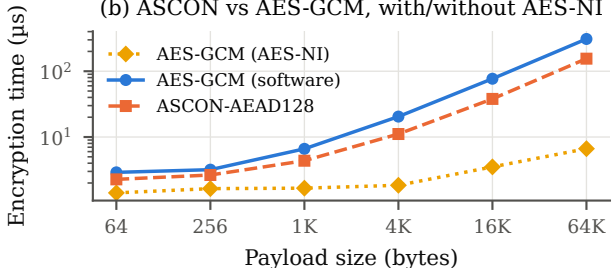

(a) Cost per mode (AES-NI gateway)
Encryption time (µs)
10^4
10^2
10^0
None
ASCON-128
AES+Mask
AES-GCM
AES+PHE
64
256
1K
4K
16K
64K
Payload size (bytes)


(b) ASCON vs AES-GCM, with/without AES-NI
Encryption time (µs)
10^2
10^1
AES-GCM (AES-NI)
AES-GCM (software)
ASCON-AEAD128
64
256
1K
4K
16K
64K
Payload size (bytes)